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Roll No

EC-8001-CBGS

B.E. VIII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

VLSI Design

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Assume suitable data, if required.

1. a) Draw and Explain Twin Tub process in detail.
b) Explain Latch up triggering and Internal Latch up prevention techniques.
2. a) Write and explain the classification of integrated circuits by device count.
b) Explain the microelectronics field. Give the types of major processes used in IC fabrication.
3. a) Draw and explain the block diagram of conventional IC design process.
b) Explain the operating principle of n channel MOSFET with the help of suitable regional diagrams.
4. a) Write the difference between Bipolar technology and Hybrid technology.
b) With neat sketches explain oxidation process in IC fabrication.

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5.
 - a) Explain the principle of Quasi static register cells.
 - b) Write difference between Random Logic and structured logic forms.
6.
 - a) Explain level-2 large signal model and compare it with the high frequency model.
 - b) What is the need for design rules? Explain different types of design rules.
7.
 - a) Explain the operation of micro coded controller with neat sketches.
 - b) What do you mean by systolic Arrays? Explain its multiplication process.
8.
 - a) Explain different steps involving in n well CMOs process.
 - b) Discuss the physical origin of latch up and discuss some latch up prevention techniques.
